Supplementary Materials

Validation-first measurement of resilience in illicit data economies

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The Supplementary Materials contain the complete audit trail; every number in the main text can be traced to a note below. No number, claim or citation is introduced here that is not already in the main text or in the source analysis records.

S1. Complete protocol detail

S1.1 The two-list rule

When an instrument self-reports failure, the failure report itself is that gate's output and must be given: how many cells fell outside the window, which cells they were, what the window was, and the verbatim text of the decision rule. What is forbidden is any quantity *read off* the failed instrument — a point estimate, a band, a comparison, a significance statement. The test for which list a reading belongs to is whether it can stand without the failed instrument: cell identities and the window cannot, and are therefore reportable; a shuffled-control AUC can, and is therefore withheld.

S1.2 What would falsify the protocol

A protocol that cannot fail is not one. Three observations would count against it, and each is checkable. (i) If, in a new domain, the G2 controls pass for *nearly every* candidate instrument, then G2 has no discriminating power there and the gate is a formality. (ii) If the metrics G1 screens out bear no relation to the metrics practitioners in that domain consider valid, G1 is pedantry rather than protection. (iii) If the audit trail never changes a reported number, and no quantity is ever withheld, downgraded or reversed by the controls it is paired with, then G3 is bookkeeping. Our own corpus is a partial answer in the opposite direction: the controls reversed four of our claims, three instruments failed outright, and no metric cleared all three validity gates, so the gates bit on this domain.

S1.3 The audit census

The manifest has a time point and a domain, and both matter. Its domain is the result artifacts this study produced; its time point is the manifest's build, and artifacts produced after it are not in it. The artifacts counted are **302 in all**, distributed across the top-level tree (196), the figures subtree (25), and eight further subtree groups (40, 14, 9, 5, 5, 3, 3 and 2). Earlier drafts reported **283** entries and flagged six de-leakage artifacts carried outside them; that gap is closed here rather than carried forward, and no artifact in the manifest's scope is unregistered. A later build of the same tree added the eight outputs of the four pre-registered L4 cells — one JSON and one plain-text result file per cell — and the totals above are those of that build. Because a census is only as good as the rule that draws its boundary, that rule — which trees are walked and what counts as an artifact — is reproduced with the manifest, so that the count can be recomputed rather than taken on trust.

S1.4 Reproduction discipline

Cross-artifact agreement is required before a quantity may anchor an analysis: two pipelines independently reproduced an address count of **61,487**. CPU and GPU replicates are never tabulated together; on one replicate the largest per-cell absolute difference in reported AUC was max $|d| = 0.0068$ (cell 5/Q4), against a pre-registered tolerance of 1e-3. Device-equivalence checks under fp64 deterministic mode reproduce bitwise across devices; fp32 forward passes exhibit chaos amplification and are excluded from equivalence gates.

S2. Full controls, rejected estimators, specification curve and metric validity

S2.1 Full controls and every rejected estimator

Every control reported in the main text, with the estimator it rejected, is listed in the analysis table of the main text. Three entries carry additional detail. For E7-inf, a second apparent effect (significant survival differences across groups) reduces to a label-semantics difference between untagged placeholder and source-probe groups and is disclosed as such; and one control is vacuous by construction, since hosts carrying a single URL cannot produce a later addition and both kinds read exactly 0.00% against approximately 8% floors. For E0, the concentration proposition fails under clustered standard errors (**p = 0.1736** by reporter, **p = 0.5978** by block) after the IID standard error understated the block-clustered one by **14.44**×. For E5-fin, every p-value is withheld: between-step differences span −27.4 pp to +13.1 pp, roughly 100× the nominal standard error, so the fixed-effect standard error assumes a homogeneity the data violates.

S2.2 Specification curve (E1-v3)

The demoted estimate's specification curve spans twelve core specifications that vary the fixed-effect structure, the binning, the event definition, the target market and the pre/post window. All twelve are estimable, every one is positive and clears zero at the 5% level, β spans **3.2892 to 4.1176** with median **3.7755**, and the positive share is **1.0**. The curve is anchored to the incumbent analysis rather than standing free: with the candidate set held equal it reproduces that analysis's coefficient exactly ($|\Delta\beta|$ = **0.00e+00**), and on its own candidate set the gap is $|\Delta\beta|$ = **0.0360** (**3.6005** against **3.6365**) over **489** candidates of which **158** are exposed. The five auxiliary specifications span **2.2926 to 4.2076**, and one of them does not clear zero (**0.1468**, p = 0.2723). The corrected magnitudes, **+1.0424** and **+2.4426**, come from designs whose nulls sit near zero and lie below this range rather than inside it. The curve varies the specification, not the placebo machinery, so it does not re-test the parallel-trends violation, and its breadth is not an interval estimate.

S2.3 Exploratory cross-layer association (WP4)

Thirteen usable months; the association between the social layer's monthly feedback volume and the financial layer's monthly transaction volume is read as a lag profile over lags −3 to +3. The same-period association is **r = -0.7206**, the largest absolute correlation in the window, with the next-largest at lag +1 ($|r| = 0.6867$). The circular-shift null (10,000 draws) degenerates on a window-maximum statistic: the profile maximum is deterministic under shift, so the draws take only **4 distinct values** and **50.4%** of them fall on the observed value itself; the 95% interval collapses to [**0.5365, 0.7206**] and the in-artifact **p = 0.5038** is a tautology rather than a test. We therefore claim nothing about whether the layers are coupled. This analysis is exploratory and is not one of the seven controlled analyses.

S2.4 Metric-validity matrix

Six candidate resilience metrics were screened against three pre-registered criteria: rank recovery against a planted truth without collapsing onto another metric's truth, calibration whose slope interval covers 1 and intercept interval covers 0, and separation from a week-permuted null. **Each criterion passes two of the six metrics, and no metric passes all three**; two metrics pass two of the three. The metric with the highest own-truth correlation (`recovery_time`, Spearman **0.9859**) fails the rank criterion because it is indistinguishable from `centrality_redistribution` (max |cross-Spearman| = **0.8407** $\geq$ 0.8), and the two metrics that pass the rank criterion fail the permutation criterion by construction. The artifact lists the three criteria without stating a conjunction rule, so any single count of survivors is a post-hoc choice; we report the per-criterion matrix. The six metric *names* come from the pre-registered plan while their operational definitions are ours, so none of this validates the delivered E1 measure.

S3. Per-configuration anatomy of the negative results

S3.1 De-leakage configurations (E5p/E5r)

Read on the within-entry-step metric the arms are judged on: configuration L1 keeps nine of the as-built arm's ten features (all but the calendar column) and reads **0.7155632**, *above* the as-built baseline of **0.7136417**; configuration L2 keeps three features and reads **0.6560979**; configuration L3 reads exactly **0.5000** and the artifact marks it at chance, but L3 retains only the stratifying variable itself, and a single feature constant within every bin yields an AUC of exactly one half by construction, so L3 is a vacuous control and its 0.5 is not evidence of repair. **The strongest non-vacuous configuration moves the control from 0.7136 to 0.6561 and stops there.** The values 0.633 / 0.652 / 0.681 that the same artifact reports are its *pooled* per-horizon figures, which the artifact itself notes are not comparable to 0.5. One limitation is declared inside the artifact: the de-leakage removes node and tabular feature columns, but graph message passing still runs over the full snapshot graph. An untrained model of the same architecture already separates at up to **0.6378** (GraphSAGE, $\Delta = 1$), and across the three configurations the untrained GraphSAGE readings span **0.3672**–**0.6378**, so any AUC inside that band is architecture rather than learning. The pre-registered feature matrix contains a column that *is* the $\Delta = 1$ label (`num_timesteps_appeared_in` ≥ 2): a baseline consuming it reaches test AUC **1.0**, and the artifact declares the same baseline inadmissible in the same breath.

S3.2 Changepoint detection (E14)

Unsupervised changepoint detection on temporal embeddings of the market graphs missed every reference event: the pre-registered gate reads a **firing rate of 0/25 = 0.000** and clears its own threshold (≤ 0.20) by missing rather than by detecting. Three baselines make the zero interpretable: the shifted-date control also fires 0/6 on the Agora substitution and 0/4 on the Evolution substitution; a second data source fires 0/16; and a volume-only control *does* fire (rate 0.040, detecting 2017-01), so the detector is not inert. The originally specified events (AlphaBay/Hansa, 2017-07) lie *outside* the data's time range and were not tested, and the events actually used, Agora 2015-08 and Evolution 2015-03, are substitutions. We read the 0/25 as a power result rather than merely a miss: at this graph scale and embedding fidelity, effects the size these seizures are assumed to have lie below the detection floor.

S3.3 Three dead ends

Two-event consistency (E2) is infeasible by pre-registered criterion: no acquired dataset brackets either Operation Onymous (2014-11) or Bayonet (2017-07), because the AlphaBay archive ends

2017-01-28, roughly six months *before* Bayonet, and Onymous predates the acquired coverage. Role annotation (E10) never reached data acquisition; the role-labelled pipeline does not exist and no number in the paper depends on it. External validation against monthly breach-posture reports (EV-legacy) was rejected as a validation target: per-issue filtering windows drift, one issue's window covers the previous month, and two internal series disagree during the anchor's most volatile months.

S4. Family-level profile and scenario robustness

S4.1 Family-level infrastructure profile (WP8)

This is a family-level profile reported as description only. Among the **8** malware families with at least **200** listed rows in the E0 cohort — `mozi`, `mirai`, `(unknown)`, `luajit-loader`, `ua-wget`, `botnetdomain`, `clearfake` and `censys` — the week-7 reconstitution rate spans **66.582 percentage points** from the lowest family to the highest. No test of family-level differences was pre-registered, so we give no p-value and make no claim that any family reconstitutes faster or slower than another. The `(unknown)` entry is reported under its own name, because it is both a family bucket and a missing-value bucket and renaming or merging it would let a missing-value bucket read as a real family.

S4.2 Scenario robustness (E8)

The data-driven scenario analysis of intervention rankings replicates exactly (bit-level agreement with the published target) and its own replication gate self-reports as passed, but its rankings are stable in only one of four scenario families (27/30 resamples in S3; 3/30 in S4; 0/30 in S1 and S2), and its cascade amplifier never fires at the headline threshold (amplification 0.73–0.77 < 1 at the stated threshold). Its sibling in the same family is worse: the calibration's achieved coefficient is **-1583.33** against a target of **2.4426**, so the sign is reversed and the magnitude is off by approximately 648×; all four of its scenarios are null and its bootstrap count is 0. We therefore report **no intervention ranking as a conclusion**; the scenario machinery is included to show what stability would have to look like.

S4.3 The E0 infrastructure-survival panel

The six-panel figure that carries the E0 survival result of §4 of the main text is reproduced here, with its caption as it stands in the source. It is the one main-text figure moved out of the body for space rather than for content: E0 is not one of the three findings that survive, and its headline number (median lifetime **1.36** days) is carried in the main text by Table 2 and by §6.

S5. Data, time coverage, licence and ethics supplement

S5.1 Per-source time coverage

The three layers do not share a time axis. The AlphaBay full-site archive is right-censored at a single instant (2017-01-28) and carries no seller identifier, which is why the social-layer unit of analysis is the listing rather than the vendor. Agora spans 2014–2015. The Kilos cross-market review panels and the drug-listing corpus carry no date column at all, so no window can be stated for them. The Elliptic++ time axis is **ordinal steps**, not calendar dates; it can nevertheless be anchored from outside the data, because the steps span blocks 391,200–487,975 at 2,016 blocks per step, so block 400,000 = 2016-02-25 placed against 144 blocks/day maps the window to approximately 2016-01-01 to 2017-11-03 and puts the July 2017 enforcement wave at steps 41–43. The map is approximate to ±1 step (roughly two weeks); it is the only calendar anchor available in the financial layer, and

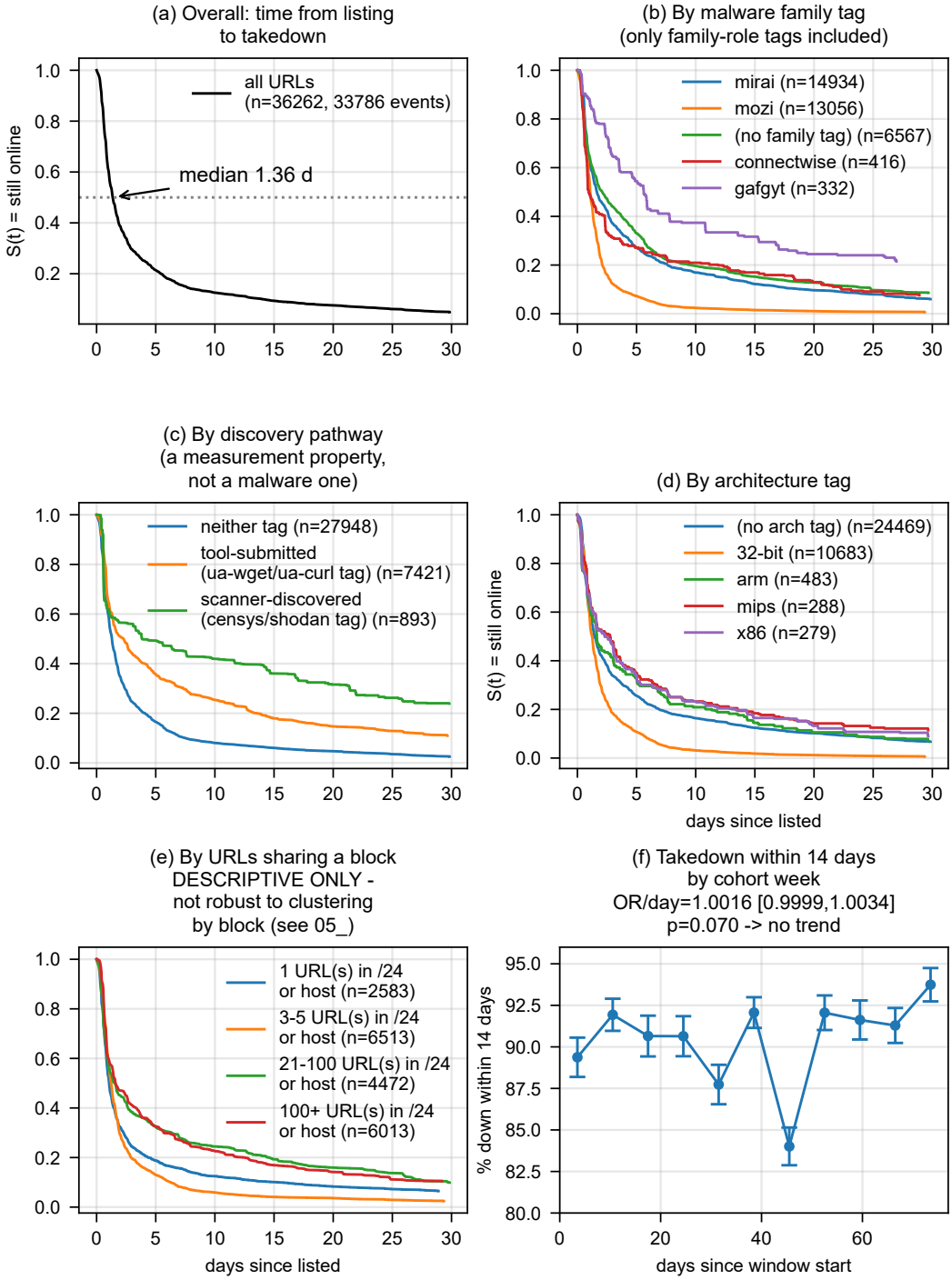


Figure S1: Survival of listed malicious infrastructure (E0). Kaplan–Meier survival over 36,262 URLhaus records listed between 2026-06-25 and 2026-09-23 (89 days), 33,786 events and 2,476 censored entries. Every duration is a lower bound: `last_online` records the last sighting of a URL, not the instant it was taken down. (a) All URLs. (b) By malware family tag; only family-role tags are included. (c) By discovery pathway, which is a property of the measurement route rather than of the malware. (d) By architecture tag. (e) By how many URLs share a /24 block or a host; this panel is descriptive only and is not robust to clustering by block. (f) Takedown within 14 days by cohort week, with the fitted logistic trend OR/day $= 1.0016$ $[0.9999, 1.0034]$, $p = 0.070$, i.e. no trend. Source: URLhaus; audited result artifact available from the corresponding author.

nothing else in the paper depends on it. The URLhaus and ThreatFox feeds are rolling-retention: an old URL survives in the dump only while it still resolves, so removal times exist only inside the recent window.

S5.2 Additional and context sources

Two further social-layer sources were acquired and probed for the cross-layer-bridging hypothesis that E3 reports as unmeasurable: the Hansa and Valhalla dumps (51,553 reviews and 16,511 listings). No reported quantity is drawn from them. Nine monthly global breach-posture reports (2025-01 through 2025-11, with 05 and 09 absent) are used only as qualitative motivation, because their per-issue filtering windows drift and the 2025-10 issue's window covers September; they are not used as evidence.

S5.3 Licence and ethics detail

We use metadata and public archives only; no raw leaked personal records are touched; identifiers are hashed. Eight sources are public archives used under research terms. Two Gwern-curated archives carry no explicit licence and are treated as research fair use and flagged for the community; the Abuse.ch feeds are free for research with attribution requested; the Elliptic++ datasets are public datasets for research use. The nine breach reports' known window defects are disclosed rather than discovered by a reviewer later. LLM-based tooling was used in exploratory analysis only, and no reported number depends on LLM-generated labels, because that pipeline never reached data acquisition. An early packaging artifact contained a default service credential; it was rotated, purged from all packaged snapshots, and the incident is disclosed rather than left discoverable.